

DDS $\longleftrightarrow$ Arc Space: Correspondence to Verify

Setup

Let $K : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$ be the KL divergence (population loss), with $K(w^*) = 0$ at a trained parameter w^* . Write $W_0 = \{w : K(w) = 0\}$ for the singular set (solution locus).

Two Definitions of the Same Object

Definition 1 (DDS, Shirodkar 2026). *A unit direction $u \in \mathbb{R}^d$ is a **dead direction of KL order k** at w^* if*

$$K(w^* + tu) = ct^{2k} + O(t^{2k+1}), \quad c > 0.$$

The Fisher information decays along u as

$$u^\top F(w^* + tu) u = \Theta(t^{2(k-1)}).$$

Definition 2 (Arc spaces / contact loci). *The **m -th contact locus** of K at w^* is*

$$\text{Contact}_m(K) = \{ \gamma \in J_m(\mathbb{R}^d)$$

$\mid$
 $\text{ord}_\gamma(K) \geq m + 1 \}, \text{ where } J_m(\mathbb{R}^d) \text{ is the space of } m\text{-jets (arcs truncated to order } m\text{). A linear arc } \gamma(t) = w^* + tu \text{ satisfies } \text{ord}_\gamma(K) \geq m + 1 \text{ iff } K(w^* + tu) = O(t^{m+1}).$

Proposition 1 (The correspondence). *A direction u is a dead direction of KL order k if and only if the linear arc $\gamma(t) = w^* + tu$ lies in $\text{Contact}^{2k-1}(K)$ but not in $\text{Contact}^{2k}(K)$.*

Proof. By definition, u has KL order k iff $K(w^* + tu) = ct^{2k} + O(t^{2k+1})$ with $c > 0$. This is equivalent to $\text{ord}_\gamma(K) = 2k$, i.e. $\gamma \in \text{Contact}^{2k-1}(K) \setminus \text{Contact}^{2k}(K)$. $\square$

Two Formulas for the RLCT

Theorem 1 (DDS selection rule, Shirodkar 2026 Thm. 3). *On smooth singular fibres, a dead direction of KL order k contributes directional RLCT*

$$\lambda_u = \frac{1}{2k}.$$

The rate exponent $2(k-1)$ of Fisher decay recovers λ_u via $\lambda_u = 1/(2 + \text{rate})$.

Theorem 2 (Ein–Lazarsfeld–Mustaa, arXiv:math/0303268).

$$\lambda = d - \sup_{m \geq 0} \frac{\dim \text{Cont}^m(K)}{m + 1}$$

Correspondence Table

DDS language	Arc-space language
Dead direction of KL order k	Arc $\gamma(t) = w^* + tu$ in $\text{Contact}_{2k-1}(K)$
Fisher decay rate $2(k-1)$ along u	Contact order $2k-1$ of the arc
Selection rule: rate $\mapsto$ RLCT	Musta�a's formula: $\text{codim}(\text{Contact}_m) \mapsto \lambda$
$\lambda_{\min}(F(w^* + tu)) \rightarrow 0$ as $t \rightarrow 0$	Contact locus $\text{Contact}_m(K)$ gaining dimension

The Unifying Claim

The DDS selection rule $\lambda_u = 1/(2k)$ and Musta a's formula are the same statement: a direction where K vanishes to order $2k$ contributes a directional log canonical threshold of $1/(2k)$. DDS reads this from the Fisher spectrum at one checkpoint. Musta a reads this from the dimension of the contact locus.

Things to Check by Hand

1. Verify the Proposition directly for $K(a, b) = a^2b^2$, $w^* = 0$, direction $u = (1, 0)$: $K(tu) = 0$ for all t , so $\text{ord} = \infty$. This is a direction *inside* W_0 , not a dead direction in DDS sense (DDS requires $c > 0$). Check: does the contact locus formulation agree?
2. For direction $u = (1, 1)/\sqrt{2}$: $K(w^* + tu) = t^4/4$. KL order $k = 2$. Check: $\gamma \in \text{Contact}_3(K)$, $\gamma \notin \text{Contact}_4(K)$. Directional RLCT = $1/4$.
3. Verify Musta a's formula for $K = a^2b^2$ with $n = 2$, $m = 1$: $\text{Contact}_1(K) = \{(a_0, a_1, b_0, b_1) : a_0^2b_0^2 = 0\} = \{a_0 = 0\} \cup \{b_0 = 0\}$ in $J_1(\mathbb{R}^2) \cong \mathbb{R}^4$. Dimension = 3. Formula gives $(2 \cdot 2 - 3)/1 = 1$. But true $\lambda = 1/2$. *Where does the discrepancy come in? The formula uses $\text{lct}(K)$, and $K \geq 0$ forces $\text{RLCT} = \text{lct}(K)/2$ due to even vanishing. Check: $\text{lct}(a^2b^2) = 1$, $\text{RLCT} = 1/2$. ✓*
4. The branch-selection issue: on branch $\{b = 0\}$, the contact locus is $\{b_0 = 0\}$, codimension 1, giving $\text{lct} = 1$, $\text{RLCT} = 1/2$. On branch $\{a = 0\}$, same. At the origin where branches meet, Contact_1 has both components; the formula still gives $1/2$. Verify the DDS selection rule gives the same from each branch.